Tidal potential measured with distant optical clocks

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The Moon and the Sun change Earth's gravitational potential directly and through solid-Earth deformation and ocean motion,[1,2] altering the relative rates of distant clocks.[3] Fibre-linked optical clocks can track these changes,[4,5] but measuring tidal signals and quantitatively separating their overlapping components remain challenging. Here we detect tidal variations in the potential difference between a strontium clock[6] in coastal Shanghai and an ytterbium clock[7] in inland Wuhan, 692 km apart. Across 545.9 hours of observations spanning 353 days, the response to the combined solid-Earth and ocean-loading prediction is 0.96 ± 0.11. The lunar semidiurnal group has a response of 1.00 ± 0.13. Spectral peaks near the M_2 and S_2 frequencies occur at 22.361 and 23.148 μHz, with equivalent-height amplitudes of 4.87 and 2.87 cm, respectively. Each contribution remains identifiable after controlling for the other. We develop a signal-to-noise model that combines signal strength, clock noise and actual sampling, accounts for the lunar precision, and helps explain the weaker constraints on other components. These results help characterize the tidal background in optical-clock networks, with potential applications to the redefinition of the second[8] and the unification of height systems at the centimetre level.[9]

Introduction

Optical clocks connected by frequency-transfer links measure gravitational potential differences through the gravitational redshift. In the weak-field limit of general relativity, a potential difference ΔU produces a fractional frequency shift $\Delta\nu/\nu \simeq \Delta U/c^2$, where c is the speed of light.[3] Evaluated fractional systematic uncertainties now approach 10^{-19}.[10] Near Earth's surface, fractional shifts of 10^{-18}, 10^{-19} and 10^{-20} correspond to equivalent heights of approximately 1 cm, 1 mm and 0.1 mm.[5,11] The redshift has been resolved within a millimetre-sized atomic sample,[11] while fibre networks and free-space frequency transfer extend comparisons to distant sites.[4,12–14] Proposed quantum clock networks[15] have also reached an elementary demonstration with entangled clocks.[16]

Clock comparisons complement levelling and satellite gravity observations.[17,18] Earth's gravitational potential also changes with volcanic activity,[19] tsunamis[20] and water storage.[21] These contributions overlap in a clock record, so detecting a weak signal requires identifying the stronger variations and knowing how accurately each can be removed. Tides provide a predictable signal with which to develop this capability. Lunar and solar contributions must be distinguished from each other and from laboratory variations that follow the solar day, while observational gaps spread spectral power into neighbouring frequencies. Subtracting the tidal background therefore requires separating the tidal components and their sampling aliases and quantifying the precision of each component.

Here we report measurements of tidal changes in the potential difference between a strontium clock in coastal Shanghai[6] and an ytterbium clock in inland Wuhan,[7] 692 km apart and connected by a 1,350-km fibre link.[22] Over 545.9 hours of observations spanning 353 days, we measure the combined solid-Earth and ocean-loading tide and separately detect the principal lunar semidiurnal contribution (M_2) and the contribution at the principal solar semidiurnal frequency (S_2). The two contributions drift in relative phase and produce neighbouring spectral peaks, and each remains identifiable after the other is removed. We develop a signal-to-noise model that combines signal strength, noise and the actual sampling, and show that it accounts for most of the differences in precision between components.

Results

A coastal–inland clock baseline

As shown in Fig. 1a, the Shanghai Sr clock and the Wuhan Yb clock lie on a 692-km coastal–inland baseline and are connected through Hefei by a 1,350-km phase-stabilized fibre link.[22] The comparison instability is approximately $5.4 \times 10^{-16}/\sqrt{\tau/\mathrm{s}}$ at averaging time τ for the spliced summer record (ref. 23) and is dominated by the clocks. Changes in the frequency ratio follow the changing potential difference, and a fitted offset for each observing segment absorbs the static part.

On this baseline, ocean loading contributes appreciably. We compute the solid-Earth tide following the IERS conventions[1] and the loading potential from FES2022,[24,25] including water attraction, induced Earth potential and clock displacement.[2] For the principal lunar semidiurnal tide, M_2, the differential solid-tide and loading amplitudes are 3.41×10^{-18} and 1.42×10^{-18} in fractional frequency, and their 20° phase difference gives a combined amplitude of 4.76×10^{-18} (Fig. 1c).

The observations span five campaigns and 54 observing segments over 353 days (Fig. 1b). The main analysis uses 545.9 h of retained samples to estimate template responses, spectra and signal-to-noise ratios. A filtered analysis of the same observations provides the time-domain displays (Fig. 2) and validation tests. Both analyses fit segment offsets; their selection, weights and template realizations are described in Methods and Extended Data Table 1a.

Amplitude, phase and coverage

The full-template response is 0.96 ± 0.11, where a response of one denotes agreement with the predicted change in frequency ratio (Fig. 3a). To separate the semidiurnal contributions, we group M_2 with the larger lunar elliptic semidiurnal tide, N_2, and the principal solar semidiurnal tide, S_2, with the lunisolar semidiurnal tide, K_2. Within each group, relative amplitudes and phases follow the physical model. The lunar-group response is 1.00 ± 0.13 (Fig. 3a), and allowing its phase to vary gives timing consistent with the prediction (Fig. 3b).

The lunar and solar semidiurnal periods are approximately 12.42 h and 12 h, and their changing relative phase allows the observations to distinguish them. The two-month summer campaign covers the full relative-phase cycle, whereas the shorter campaigns cover narrower intervals (Fig. 1d).

The five filtered-analysis campaign estimates are mutually consistent (Extended Data Fig. 1a,b and Extended Data Table 2a). After removal of fitted offsets and non-lunar groups, the summer time series and its dependence on lunar phase agree well with the predicted variation (Fig. 2). The time axis compresses gaps; Extended Data Fig. 2a shows all campaigns, and panels b,c show April in calendar time.

The 274-h summer record was also used in the absolute frequency-ratio measurement of Hou et al.;[23] Extended Data Table 2b relates those estimates to the present analysis.

Separate semidiurnal detections

The equivalent-height amplitude spectrum (Fig. 3c) has lunar and solar-frequency peaks of 4.87 and 2.87 cm at 22.361 and 23.148 μHz. Subtracting the fixed unit-amplitude physical prediction reduces the amplitudes at these frequencies to 0.71 and 1.02 cm. Each principal frequency retains its peak when the other is controlled by a freely phased sinusoid, with N_2 and K_2 also controlled (Extended Data Fig. 3). A joint fit gives M_2 and S_2 amplitudes of 4.13 and 1.76 cm (4.50×10^{-18} and 1.92×10^{-18}; Extended Data Table 2c). These differ from the scan peaks because neighbouring frequencies are fitted simultaneously. The conditional spectra thus separate the M_2 and S_2-frequency contributions without imposing their astronomical phase relation. The predicted-phase group responses in Fig. 3a instead test agreement with the physical waveforms, keeping the relative phases of S_2 and K_2 fixed within their group (Methods).

Reliability of the separated signals

Sampling aliasing in the finite, interrupted record introduces correlations between frequency components, which we test by reciprocal subtraction (Extended Data Fig. 4). Removing the jointly fitted lunar waveform changes the solar-frequency amplitude from 2.67 to 1.76 cm (2.918 to 1.921×10^{-18}). Removing the solar-frequency waveform changes the lunar amplitude from 4.45 to 4.13 cm (4.857 to 4.505×10^{-18}). Both remain identifiable. The maximum canonical correlation between the projected sine–cosine pairs is 0.290 (Extended Data Fig. 5), and the strongest sampling aliases respond to lunar-waveform subtraction as predicted (Extended Data Fig. 6).

The lunar estimate is also stable against specified daily disturbances. A 10^{-18} fractional-frequency sinusoid at 24 h or 12 h changes the main lunar response by at most 0.023 with a phase fixed across the record; with independent phases in each campaign, the bounds are 0.030 and 0.12, respectively. The filtered analysis provides complementary checks of daily terms, sampling and data selection (Fig. 4). Both clocks correct blackbody shifts in real time, and temperature regressions test residual correction errors (Methods and Extended Data Fig. 1d). In the filtered analysis, adding the fibre-link monitor as a regressor changes the lunar and full-template responses by less than 0.005. Retaining the screened transients leaves the lunar result consistent, and the fits improve predictions for most withheld observation periods (Methods). Extending the ocean mask near Shanghai by 0.3° or 0.6° changes the filtered lunar response by −0.05 or −0.08, respectively, about half of its statistical uncertainty.

Ocean loading and diurnal response

The separate ocean-loading and diurnal responses are constrained less precisely, with signal-to-noise ratios near unity (Table 1). The corresponding filtered-data fits are consistent with the prediction (Extended Data Fig. 1b; Supplementary Methods SM4 and SM5). Refined coastal loading models and longer continuous runs are expected to sharpen these comparisons.[26] These weaker constraints motivate a quantitative account of component precision.

Discussion

We develop a signal-to-noise model that relates component precision to signal strength, noise and phase coverage. The observation schedule enters through the sampling window and coupling between fitted components; within short segments, a slowly varying tide is partly absorbed by the fitted offset. For a specified estimator, the amplitude signal-to-noise ratio scales locally as

$$\rho_i(T) \simeq \rho_i(T_0)\frac{\eta_i(T)}{\eta_i(T_0)}\left(\frac{T}{T_0}\right)^{a_i-b_i}. \tag{1}$$

Here T is the retained exposure and $\rho_i = S/R$. The exponents a_i and b_i describe how the phase-aligned signal and the noise standard deviation accumulate, and η_i accounts for the sampling window, weights and parameter coupling. For a coherent signal in white noise, $a_i = 1$ and $b_i = 1/2$,

giving $\rho_i \propto \sqrt{T}$ at fixed η_i. We evaluate this white-noise case on the actual sampling with the measured amplitudes and a common clock-noise coefficient (Methods). Table 1 compares the resulting precision with the empirical errors of the same estimators.

The model-estimated signal-to-noise ratio closely matches the measured value for the lunar component, whereas agreement is poorer for the weaker components (Table 1). This difference may reflect the greater relative impact of residual contamination through sampling aliases on weak signals, together with non-white laboratory variations at 24 h and 12 h that could increase the uncertainties of the diurnal and solar-frequency components (Methods; Extended Data Fig. 1d; Supplementary Methods SM4). Short segments constrain diurnal cycles weakly after offset removal, and the smaller loading signal overlaps the solid-Earth waveform. The signal-to-noise model thus accounts for most of the differences in precision between components. This comparison evaluates precision at the measured amplitudes; the exposure scaling in equation (1) remains conditional on the noise and sampling efficiency of future observations.

The model identifies distinct ways to improve future observations. Better clock stability lowers noise for a given retained exposure, whereas sustained network operation improves phase coverage and reduces component coupling. Continuous runs of at least two days with maintained phase lock cover successive diurnal cycles; a longer calendar span separates neighbouring frequencies, while gaps create sampling aliases (Extended Data Fig. 6). Increasing retained exposure accumulates signal according to equation (1), with the applicable noise and sampling efficiency. Together with additional sites and local environmental records, these improvements would allow weaker potential changes to be identified after subtraction of the tidal background.

The measurement and separation of the lunar and solar-frequency semidiurnal contributions extend demonstrated sub-centimetre-equivalent clock performance[5] to time-varying geophysical signals. Together with the signal-to-noise model, they support the development of clock networks for geodesy, ultralight-dark-matter searches[27] and future clock-based gravitational-wave detectors.[28] Characterizing the tidal background has potential applications to the redefinition of the second[8] and the unification of height systems at the centimetre level.[9]

Note added. During the preparation of this manuscript, a related observation of tidal variations of the gravity potential in a European optical-clock network was reported;[29] the present work is complementary in separating the lunar and solar semidiurnal contributions on a coastal–inland baseline.

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Table 1 | Model-estimated and measured signal-to-noise ratios.

Evaluation	Lunar M_2	Solar S_2	Diurnal group	Ocean loading
S/R (model estimated)	7.17	2.78	1.55	1.17
S/R (measured)	7.02	1.96	1.08	0.89

$S = |\hat{A}|$ is the measured amplitude or template-response magnitude and $R = u(\hat{A})$ its standard error. The model-estimated row assumes white frequency noise and uses the common noise coefficient and the actual sampling; the measured row uses UTC-day-jackknife covariance. All entries use the auxiliary reduction: M_2 and S_2 from their joint sine–cosine fit, the diurnal group from the three-group fit and ocean loading from the two-template solid-Earth/loading fit (Methods).

Main figure legends

Fig. 1 | The predicted tide on a coastal–inland baseline. a, The Sr clock in Shanghai, the Yb clock in Wuhan and the fibre link through Hefei; the dashed line marks the 692-km distance between the clocks. Land contours show the predicted M_2 tidal potential sensed by a clock, including the solid Earth tide and ocean loading, in units of 10^{-18} in fractional frequency (interval 0.5; integer contours labelled). Sea shading and contours show the FES2022 M_2 ocean-tide amplitude (interval 25 cm; 50, 100 and 150 cm labelled). The inset shows the ocean-loading contribution within the dashed box around Shanghai, with labelled contours at 1 and 2×10^{-18}. **b,** Observation calendar of the five campaigns from September 2025 to August 2026. Bars mark the retained observation intervals in the filtered validation analysis; brackets span campaigns with separated runs. **c,** Predicted differential M_2 phasors of the solid Earth tide, the ocean-loading potential and their sum, referenced to the solid Earth tide; the loading phasor is at 20° to the solid Earth tide. **d,** Relative astronomical phase of S_2 and M_2, including the nodal correction, at the filtered 600-s samples, shown as one tick per sample on a −180° to 180° axis. The short campaigns cover limited ranges of relative phase, whereas the summer campaign covers the full cycle, with some gaps, as the lunar tide moves through the solar day.

Fig. 2 | The lunar tide in the summer record. Both panels use the filtered validation analysis (Methods). **a,** Three-hour running averages of the June–August 2026 frequency ratio after subtracting fitted segment offsets and the fitted near-12-h and diurnal (non-lunar) tidal groups (blue), with the lunar model (black). The measurement-time axis compresses gaps longer than 6 h; grey lines mark their removal, and values below -13×10^{-18} are clipped (open triangles). White frequency noise alone corresponds to a scatter of approximately 5×10^{-18} for a 3-h average. **b,** The same data folded at the model M_2 phase, relative to the predicted maximum, in 30° bins, with the lunar model (M_2 and N_2) folded on identical sampling. Error bars are day-by-day jackknife uncertainties with the fitted non-lunar terms held fixed. No empirical daily terms are fitted.

Fig. 3 | Amplitude, phase and frequency of the tidal response. All panels use the main analysis (the auxiliary reduction) into 3,299 non-overlapping 600-s means with 54 segment constants. **a,** Complete-template (‘full tide’) response and jointly fitted lunar (M_2, N_2) and near-12-h (S_2, K_2) group responses, from separate fits with period-dependent noise weights. Error bars are one stan-

dard error, the larger of the scaled least-squares and UTC-day-deletion jackknife errors. **b,** Lunar-group in-phase and quadrature coefficients; the cross marks $(1, 0)$, and contours have radii one and two in the scaled least-squares covariance. The quoted marginal errors, the larger of the scaled least-squares and UTC-day-jackknife values, are 0.13 and 0.20; the scaled least-squares quadrature error is 0.127. **c,** Equivalent-height sinusoidal peak-amplitude spectra over 21–24.5 μHz, before and after subtracting the fixed unit-amplitude physical prediction, with equal weights and segment constants projected from data and trial sinusoids. The left and right axes give equivalent height and fractional-frequency amplitude, related by $A_{h,\mathrm{cm}} = 100c^2 A_y/g_0$ with $g_0 = 9.80665\ \mathrm{m/s^2}$; both are linear. Markers indicate the M_2 and S_2 frequencies; the nominal Rayleigh scale (0.03275 μHz) and grid step (0.002729 μHz) are annotated.

Fig. 4 | Robustness of the lunar amplitude. Panel a spans the two left columns, which share an amplitude range of 0.5–1.5; panel b is on the right. Both panels use the filtered validation analysis (the primary reduction). **a,** Lunar amplitude for the primary fit (filled diamond and shaded uncertainty band), alternative offset definitions, daily terms represented by air-temperature measurements or fitted 24-h and 12-h sinusoids, the summer campaign alone, omission of each campaign, and retention of segment ends or transient excursions. Error bars and the shaded band indicate one standard deviation. The thin bar below the primary result is the 95% interval from resampling operating groups. These estimates share observations. The grey interval is obtained without the summer campaign. **b,** Change in the lunar amplitude caused by a 24-h or 12-h sinusoid of fractional-frequency amplitude 10^{-18}, with one phase common to all campaigns or with the combination of independent campaign phases that produces the largest change. The dashed line marks the uncertainty of the primary estimate.

Methods

Clocks, link and data

The Sr optical lattice clock is in Shanghai (31.130° N, 121.570° E, height approximately 5 m) and the Yb optical lattice clock in Wuhan (30.529° N, 114.357° E, approximately 30 m); changing the Shanghai height by 5 m changes the differential tidal model by only 4×10^{-23}. The static geopotential difference, $280.042 \pm 0.085\ \mathrm{m^2\ s^{-2}}$ from first-order levelling [23], enters the frequency

ratio as a constant absorbed by the fitted offsets. The evaluated systematic uncertainties of the Sr and Yb clocks are 9.2×10^{-19} and 1.3×10^{-18} [6,7]; the clocks, fibre link and frequency chain are described in ref. 23, and optical frequency dissemination [22,30] over field fibres of comparable length has been demonstrated.

A 1,550-nm carrier referenced to the Sr clock through a frequency comb in Shanghai is transmitted to Wuhan, where its beat against the Yb-referenced comb is counted at 1-s intervals. In 2026 the beat frequency is 33.6 MHz, and a deviation of 1 Hz corresponds to 5.17×10^{-15} in Yb/Sr; in 2025 the counter recorded the beat after division by two, which is accounted for. The white-frequency-noise level of the tide-corrected ratio, $\sigma_y\sqrt{\tau}$, from the overlapping Allan deviation [31] at 256–4,096 s, is 7.9×10^{-16} in September–October 2025 and 4.2–5.3×10^{-16} in the 2026 campaigns, evaluated for each campaign in the primary reduction.

The 1,350-km link extends a 2,067-km dissemination system with bias-free noise compensation based on digital radio-frequency phase recording [22,23]. The out-of-loop round-trip signal and the optical phase-locked loops of the frequency chain are monitored continuously [23]; maintaining lock preserves the phase connection needed for coherent accumulation.

The observations cover five campaigns (September–October 2025, January 2026, April 2026, May 2026 and June–August 2026) and 54 segments defined by the experiment team; the summer segments are those of ref. 23. The overall effective observing time is 545.9 h, summed over the retained one-second samples in the main analysis. There are 38 segments before June and 16 in June–August. Campaigns define the seasonal subsets and campaign-specific daily terms, the eight observation periods are the units of held-out prediction, the 29 operating groups are the units of group-bootstrap resampling, and segments carry the fitted offsets. Time stamps are in local time (UTC+8).

Main analysis: grouped responses

The main analysis (auxiliary reduction) provides summary responses, spectra and signal-to-noise ratios; the filtered validation analysis (primary reduction) provides time-domain displays and robustness checks. Their observations are shared, so they are not independent measurements (Extended Data Table 1a).

Figure 3 and Extended Data Figs. 3–6 use $N_{\mathrm{b}} = 3,299$ non-overlapping 600-s means over a calendar span of 353.39 days and 54 segment constants. Within each predefined observing segment, beat-frequency records differing by more than 10 Hz from the segment median were excluded. Retained records were averaged separately within each segment on a 600-s wall-clock grid in Beijing time (UTC+8), retaining bins with at least 180 nominal one-second members. Bin timestamps denote the nominal grid centres. This reduction applies neither the triangular filter nor the additional running-median transient screening used in the filtered validation analysis. Its solid-Earth template converts the IERS radial displacements into fractional frequency,

$$y_{\mathrm{solid}} = -\frac{1 + k_2 - h_2}{h_2} \frac{g(u_{\mathrm{Wuhan}} - u_{\mathrm{Shanghai}})}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794$ m/s^2, which differ from the IERS values of the filtered-validation model ($k_{22} = 0.29801$) by 0.4% in $1 + k_2 - h_2$; its loading template uses the effective-potential convolution with 18 constituents. The lunar (M_2, N_2), near-12-h (S_2, K_2) and diurnal (K_1, O_1, P_1, Q_1) groups comprise eight constituents; the remaining terms form a fixed remainder r_0. The complete-template and grouped models are $y_i = c_{s(i)} + A_T(s_{L,i} + s_{S,i} + s_{D,i} + r_{0,i}) + \epsilon_i$ and $y_i - r_{0,i} = c_{s(i)} + A_L s_{L,i} + A_S s_{S,i} + A_D s_{D,i} + \epsilon_i$. The phase-relaxed model adds a quarter-cycle-shifted lunar template while keeping the M_2/N_2 relation.

The weights are $W_{ii} = 1/\sigma_i^2$ with $\sigma_i = \alpha_{p(i)}/\sqrt{\Delta t_i}$, where $\Delta t_i = n_i \delta t$ is the retained exposure of bin i (n_i one-second members, $\delta t = 1$ s) and α_p is the white-frequency-noise coefficient of observing period p from overlapping Allan deviations at 256–4,096 s: 9.70 and 5.70×10^{-16} for September and October 2025, and 4.30–5.76×10^{-16} for the six 2026 periods. Errors are the larger of the residual-scaled least-squares and UTC-day-deletion jackknife errors, each deletion refitting all coefficients.

Spectra and constituent separation

For Fig. 3c, at each trial frequency we fit a freely phased sinusoid and the 54 segment constants to the auxiliary means with equal weights,

$$y_i = c_{s(i)} + a_f \cos[2\pi f(t_i - t_0)] + b_f \sin[2\pi f(t_i - t_0)] + \epsilon_i,$$

and plot the peak amplitude $A_y(f) = \sqrt{a_f^2 + b_f^2}$; for the residual curve, the fixed unit-amplitude physical prediction is subtracted first. The equivalent height is $A_{h,\mathrm{cm}} = 100c^2 A_y/g_0$ with $g_0 = 9.80665\ \mathrm{m/s^2}$. The span $T_{\mathrm{span}} = 353.3889$ d sets the nominal Rayleigh scale $1/T_{\mathrm{span}} = 0.03275$ μHz. The scan spacing is $1/(12T_{\mathrm{span}}) = 0.002729$ μHz, with the astronomical frequencies added, and the peaks are local maxima within one Rayleigh scale of each prescribed frequency.

For Extended Data Figs. 3–6, a nuisance matrix N (segment constants, and free N_2 and K_2 pairs where stated) is projected out of the data and of each trial cosine–sine pair X_f with $R_N = I - QQ^{\mathsf{T}}$, where Q is an orthonormal basis of N. The fitted coefficients $\hat{\boldsymbol{\beta}}_f = (R_N X_f)^{+} R_N y$, with y in units of 10^{-18}, give the equivalent-height peak amplitude $A_{h,\mathrm{cm}}(f) = (c^2/g_0)\, 10^{-16}\, \|\hat{\boldsymbol{\beta}}_f\|$, as in Fig. 3c. At a frequency coinciding with a controlled constituent, the trial pair lies in the nuisance space and its amplitude is undefined; these entries are not plotted. Phases refer to 6 September 2025, 16:05 UTC. The M_2 and S_2 waveforms come from a simultaneous four-coefficient fit with no amplitude or phase constraint between them; without N_2/K_2 control their amplitudes are 4.63 and 1.82×10^{-18}. Conditional scans add the other principal pair to N and scan within ±0.16 μHz of the target. Compensation tests subtract a scaled donor waveform and re-estimate the target with the same nuisance terms, and canonical correlations are the singular values of $Q_M^{\mathsf{T}} Q_S$ for the residualized pairs (Supplementary Methods SM7).

Signal-to-noise ratios

We define $\rho_i = |\hat{A}_i|/u(\hat{A}_i)$. For the white-noise case of equation (1), $H = (X^{\mathsf{T}} W X)^{-1} X^{\mathsf{T}} W$ maps the observations to the fitted coefficients for design matrix X and weights W. For white frequency noise with $\alpha = 5.4 \times 10^{-16}\sqrt{\mathrm{s}}$, the bin variance is $\alpha^2/\Delta t_k$, giving

$$u_{i,\mathrm{white}}^2 = \alpha^2 \sum_k \frac{H_{ik}^2}{\Delta t_k}, \qquad \rho_{i,\mathrm{white}} = \frac{|\hat{A}_i|}{u_{i,\mathrm{white}}}. \tag{2}$$

With the retained exposure $T = \sum_{k=1}^{N_\mathrm{b}} \Delta t_k$ of the N_b means, $s_{i,\mathrm{rms}}^2 = \sum_k \Delta t_k\, s_i(t_k)^2/T$ for template s_i per unit response and $\eta_i = \alpha/(s_{i,\mathrm{rms}}\sqrt{T}\, u_{i,\mathrm{white}})$,

$$\rho_{i,\mathrm{white}} = \frac{|\hat{A}_i| s_{i,\mathrm{rms}}}{\alpha}\,\eta_i\sqrt{T}. \tag{3}$$

The sampling window, offsets, template coupling and weights enter η_i, and equation (3) gives the $a_i = 1$, $b_i = 1/2$ sensitivity at the measured amplitude (Extended Data Table 2d). For M_2 and S_2 in Table 1, $\hat{A} = (\beta_c^2 + \beta_s^2)^{1/2}$ from the equal-weight joint fit of M_2, S_2, N_2 and K_2 pairs with 54 segment offsets, and $u^2(\hat{A}) = JC_iJ^{\mathsf{T}}$ with $J = (\beta_c, \beta_s)/\hat{A}$; $C_\beta = H\Sigma H^{\mathsf{T}}$ with $\Sigma_{kk} = \alpha^2/\Delta t_k$ for the white-noise row, and the covariance of 43 UTC-day deletion fits for the measured row. The diurnal column uses the three-group fit. The loading column uses a fit of the complete solid-Earth and loading templates with two free coefficients, 54 segment offsets and equal weights, with UTC-day-deletion errors (Extended Data Table 2d and Supplementary Methods SM6).

Filtered validation: data selection

Every segment is processed identically. The time axis is reconstructed from the first time stamp and the sample index. Samples more than 10 Hz (approximately 5×10^{-14} in Yb/Sr) from the segment median are removed, the longest uninterrupted run is retained, and leading and trailing samples deviating from the median by more than 1% of the peak-to-peak range are trimmed. To remove smaller transient excursions, 300-s averages are compared with a running median over ±30 min, and averages deviating by more than six robust standard deviations are removed together with their neighbours. This step removes 0.4% of the samples, or 2.2 h (Extended Data Fig. 7d–f). The criterion uses no tidal model, and adding the model to the data leaves the selection unchanged; because the subsequent filter is linear, injection tests use the selected data. Screening variants are given in Supplementary Methods SM2.

Filtered validation: tidal model

The solid Earth contribution at each clock is

$$\delta W = \sum_m (1 + k_{2m}) V_{2m} + (1 + k_3) V_3 - g u_r,$$

where V_{2m} are the degree-2 tide-generating potentials of the Moon and Sun of order m, V_3 is the lunar degree-3 potential and u_r is the radial displacement. Love numbers and displacements follow the IERS 2010 conventions [1] (Supplementary Methods SM1). The ocean-loading potential is computed by grid convolution,

$$\delta W_{\text{ocean}}(t) = \sum_{\text{ocean}} \rho_w \zeta(\mathbf{r}', t) G_W(\psi)\, \mathrm{d}A, \qquad G_W(\psi) = \frac{ga}{M_E} \sum_n (1 + k'_n - h'_n) P_n(\cos\psi),$$

with the FES2022 ocean elevation ζ on a $1/30^\circ$ grid [24,25], $\rho_w = 1,025\ \mathrm{kg\,m^{-3}}$, Earth's radius a and mass M_E, and angular distance ψ. The Green's function [2] uses PREM load Love numbers [32,33]. After removal of the ocean mean, eight constituents (M_2, S_2, N_2, K_2 and the diurnal K_1, O_1, P_1, Q_1) are synthesized with astronomical arguments and nodal corrections. They explain 99.84% of the model variance; the remainder, with an r.m.s. of 0.18×10^{-18}, is held at its predicted value. The radial displacement from the same convolution agrees with the FES2022 load-tide grid to within 2.2% in amplitude and 1.2° in phase for seven of the eight constituents; for S_2 the grid derives from a different ocean-model release, and substituting it changes the full-template response by 0.001 (Extended Data Fig. 8; Supplementary Methods SM1). Extending the ocean mask near Shanghai by 0.3° or 0.6° changes the lunar response by −0.05 or −0.08, respectively, about half of its statistical uncertainty of 0.14.

Filtered validation: uncertainties

The 1-s observations and model are filtered identically with a 1,200-s triangular window stepped by 600 s and fitted by least squares as $y(t) = c_i + As(t)$, with an offset c_i for segment i. For band analyses, the eight-constituent model is divided into semidiurnal (M_2, S_2, N_2, K_2) and diurnal (K_1, O_1, P_1, Q_1) bands, or into lunar (M_2, N_2), near-12-h (S_2, K_2) and diurnal groups with separate amplitudes. The primary grouped fit contains no empirical daily terms. Separate fits of the solid-Earth and loading templates assess their coupling (Supplementary Methods SM4).

For the full template, the semidiurnal band and the lunar group, uncertainties are calibrated by residual resampling. The fitted residuals are shifted circularly within each continuous run of the filtered

record, added to the model with $A = 1$ and refitted; the standard deviation over 200 realizations gives the error, retaining the within-run residual structure. For the diurnal and near-12-h groups, fits with empirical daily terms and fitted daily amplitudes, we quote the larger of the scaled least-squares standard error[34] and the day-by-day jackknife error; in the resampled data these differ from the calibrated uncertainty by factors of 0.8–1.3. Group-bootstrap intervals draw the 29 operating groups with replacement in 1,000 refits; the segment bootstrap draws the 54 segments. In the phase test, each band has in-phase and quadrature templates, the latter shifting every constituent by 90°, with segment-bootstrap uncertainties. Quoted uncertainties are one standard deviation (Supplementary Methods SM3).

For Fig. 2b, data minus fitted offsets and non-lunar groups, and the M_2+N_2 model, are averaged in 30° M_2-phase bins on identical sampling. Day-by-day jackknife errors hold non-lunar fits fixed. Time-series displays use 3-h averages with marked compression of gaps longer than 6 h.

The full-record and April spectra (Extended Data Fig. 7c and Extended Data Fig. 2c) show sinusoidal power, the squared amplitude divided by two, after projection of segment offsets and before and after subtraction of the fitted whole template. Their thresholds are 99% quantiles of the maximum over 5–40 μHz from 200 resampled residual series (full record) and over 15–50 μHz from 1,000 series (April, together with pointwise 99% levels). No April frequency exceeds the latter threshold after tidal correction.

Daily terms, environment and prediction

In the filtered validation analysis, daily sensitivity fits add $a_1 \cos\omega t + b_1 \sin\omega t + a_2 \cos 2\omega t + b_2 \sin 2\omega t$, where $\omega = 2\pi/(1\ \mathrm{day})$ and t is local time, with common or campaign-specific coefficients. Alternatives use, in each campaign, a free-amplitude 24-h term with the phase of the 24-h harmonic of hourly ERA5 2-m air temperature at the two sites [35], or the air-temperature anomalies at both sites relative to a 24-h running mean as two regressors.

The response of the lunar estimate to a disturbance $B\cos(\omega t - \varphi)$ follows from the lunar row h of the estimator H (defined in the signal-to-noise calculation) for the specified fit. For a common phase, the maximum amplitude change is

$$B\sqrt{(h\cdot\cos\omega t)^2+(h\cdot\sin\omega t)^2};$$

for phases independent between campaigns, it is the sum of the corresponding campaign norms. The main-analysis bounds reported in Results use the three-group design, with the main sample times, lunar, near-12-h and diurnal templates, segment offsets and observing-period-noise weights defined in Methods. Figure 4 uses the corresponding operator for the equal-weight filtered validation fit. The following injection, environmental and held-out prediction checks use the filtered record.

Injection tests add either the full tidal model ($A = 1$) or the model without its lunar group ($A = 0$) to circularly shifted residuals of the full-template fit, under four daily-disturbance scenarios: none; campaign-specific 24-h terms with air-temperature phases and fitted amplitudes; common 24-h and 12-h terms with fitted amplitudes and phases; and the fitted summer 24-h and 12-h terms. With 300 realizations per scenario and injected amplitude, the lunar-amplitude biases lie between −0.03 and 0.00, and for $A = 1$ the fitted amplitude exceeds three standard errors in every realization (Extended Data Table 1b).

Campaign information shares project the lunar template onto the complement of the near-12-h and diurnal templates and segment offsets, and sum the squared projection within each campaign. Held-out prediction fits the amplitude to all but one observation period and applies it to the excluded period, refitting only that period's offsets (Supplementary Methods SM4).

The round-trip beat of the fibre link, recorded in Shanghai near 10 MHz at 1-s resolution, is normalized by the optical carrier frequency and filtered like the clock ratio. It covers 474 h of the analysed data. Including it as an additional regressor, or excluding 300-s blocks in which it deviates by more than 5×10^{-16}, changes the lunar and whole-template responses by less than 0.005.

The Sr vacuum-chamber temperature, the mean of 20 PT100 sensors sampled every 20 s, is available from 28 June to 7 July 2026 and overlaps 98 h of the filtered summer record. The blackbody sensitivity is 6.26×10^{-17} K^{-1} at 294.3 K. Because both clocks correct blackbody shifts in real time, a regression of the ratio on the chamber-temperature shift tests residual correction errors and has an expected coefficient of zero; it gives 0.92 ± 0.74 (Supplementary Methods SM4).

Use of large language models

Large language models were used to assist with code writing, manuscript drafting and editing. All scientific content, analyses and conclusions are those of the authors, who independently verified all numerical results and take full responsibility for the manuscript.

Data availability

The frequency-ratio records, the segment table with the retained samples and the tidal-model time series supporting the findings of this study are available from the corresponding author upon reasonable request.

Code availability

The analysis code for the primary and auxiliary reductions is available from the corresponding author upon reasonable request.

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Competing interests

The authors declare no competing interests.

Additional information

Supplementary information is available for this paper.

Extended Data legends

Extended Data Fig. 1 | Campaign-resolved responses and phase folds. a, Full-model and semidiurnal amplitude factors for each campaign and the combined observations, fitted with segment offsets. Combined-data estimates also include the lunar group and the solid Earth tide alone. **b,** Diurnal amplitude fitted jointly with the semidiurnal band in a two-band fit; the combined estimate is 0.36 ± 0.50. The arrow marks an uncertainty extending beyond the plotted range. Error bars represent one

standard deviation, calibrated by residual resampling for the full model, semidiurnal band and lunar group; for the diurnal band and solid Earth tide alone they are the larger of the scaled least-squares and day-by-day jackknife uncertainties. Thin lines in the combined-data rows show 95% intervals from resampling operating groups, or segments for the lunar group. **c,** All observations folded at the astronomical M_2 phase after subtracting fitted segment offsets and all tidal constituents except M_2 from the full-model fit. Points are means in 30° bins with day-by-day jackknife uncertainties, with the removed terms held fixed; the curve is the M_2 model. **d,** Tidally corrected residuals folded by local time in 2-h bins for summer, May (light symbols), the other three campaigns, and summer observations overlapping the temperature record from 28 June to 7 July (open circles). Error bars are day-by-day jackknife uncertainties; bins covering fewer than three days are omitted. The dashed curve is the mean daily blackbody-radiation (BBR) shift computed from the Sr chamber temperature over the same interval, after removing a 24-h running mean and excluding the temperature transient on 30 June; the real-time correction removes this shift from the recorded frequency ratio.

Extended Data Fig. 2 | Tidal records across campaigns and the April campaign. a, Three-hour averages of the filtered frequency ratio after subtraction of fitted segment offsets (blue), with the tidal model (red). Measurement-time axes compress gaps longer than 6 h, marked by vertical lines; gaps longer than 5 days are labelled. The vertical range is $\pm 15 \times 10^{-18}$; triangles mark values outside this range. **b,** Frequency-ratio observations after the 1,200-s triangular filter and removal of fitted segment offsets, their 3-h averages, and predictions with and without ocean loading. The horizontal axis retains actual calendar dates and gaps (local time, UTC+8). **c,** Sinusoidal power before and after subtracting the model, expressed as amplitude squared divided by two on a linear vertical axis, in units of 10^{-36}. The dashed curve is the 99% level at each frequency; the dotted line is the 99% level for the maximum over 15–50 μHz, both obtained from 1,000 resampled residual series. In the April record, K_1, P_1 and O_1 merge into a diurnal band and M_2, S_2 and N_2 into a semidiurnal band. Power below 8 μHz exceeds the displayed range.

Extended Data Fig. 3 | Amplitude spectra and sampling-response checks. All panels use the auxiliary reduction with the 3,299 recorded bin-centre times and their gaps, equal weights and 54 segment constants. Ordinates give equivalent-height peak amplitude in cm on linear scales, with 0–6 cm zooms below each panel. **a,** Observed amplitude spectrum and amplitude spectrum after

subtraction of the fixed unit-amplitude physical prediction, as in Fig. 3c. **b,** Amplitude spectra before and after subtraction of the fitted M_2, S_2 or both waveforms, obtained from one simultaneous freely phased fit with N_2 and K_2 controlled. **c,** Local scans around M_2 conditioned on S_2, and around S_2 conditioned on M_2, with N_2 and K_2 controlled; grey is the baseline from b. **d,** M_2-only, S_2-only and combined fitted-waveform injections through the same sampling and projection, compared with the observed spectrum; injected amplitudes need not add. Dotted lines mark N_2, M_2, S_2 and K_2 in ascending frequency. Amplitudes at the controlled N_2 and K_2 frequencies are undefined, and responses close to them are weakly constrained.

Extended Data Fig. 4 | Amplitude and phase sensitivity to compensation. a,c, Subtracting a scaled M_2 waveform and re-estimating S_2; **b,d,** the reverse. **a,b,** Equivalent-height peak amplitude, converted as in Fig. 3c; **c,d,** phase relative to the common time origin. Solid curves use donor waveforms from the joint fit; dotted curves use single-frequency donor fits. All target-only fits retain segment offsets and free N_2/K_2 pairs. Dashed grey curves (donor pair also fitted) are invariant by construction. $\lambda = 1$ denotes subtraction of the estimated waveform. Amplitude axes are linear and share a scale across directions.

Extended Data Fig. 5 | Harmonic geometry and joint-coefficient correlations on the auxiliary 600-s record. a, Correlations of the M_2 and S_2 cosine and sine design columns after projecting out 54 segment constants and free N_2/K_2 pairs. b, Normalized lag correlations of the fitted templates, evaluated on the original timestamps without interpolating missing observations; these describe template similarity, not propagation delay. c,d, Correlations between the four jointly estimated coefficients (M_2 cosine, M_2 sine, S_2 cosine, S_2 sine) from the scaled least-squares covariance (c) and the UTC-day delete-one jackknife covariance (d). The three matrices share a −1 to +1 colour scale. The maximum canonical correlation of the projected M_2 and S_2 pairs is 0.290.

Extended Data Fig. 6 | Sampling aliases and amplitude response to compensation. a,b, Equivalent-height amplitude spectra of the observations, of the observations after subtraction of the jointly fitted donor waveform, and of that donor alone, on identical timestamps and nuisance projection, with 0–6 cm zooms below. Coloured vertical lines mark three off-centre candidate frequencies selected from the pure-injection response. **c,d,** Observed amplitude at those frequencies as the compensation factor varies. Amplitude axes are linear and share scales within rows.

Amplitudes at the controlled N_2 and K_2 frequencies are undefined, and responses close to them are weakly constrained.

Extended Data Fig. 7 | Filtered-validation responses, spectra and sample selection. a, Amplitude factors for the full model and the semidiurnal band, each fitted with one amplitude, and for the lunar (M_2, N_2), near-12-h (S_2, K_2) and diurnal groups fitted jointly. All fits include segment offsets and omit empirical daily terms. Error bars are one standard deviation, calibrated by residual resampling for the full model, semidiurnal band and lunar group; for the near-12-h and diurnal groups, the larger of the scaled least-squares and day-by-day jackknife uncertainties is used. The dashed line denotes the prediction, $A = 1$. **b,** Covariance contours at radii one and two for the lunar and near-12-h amplitudes, using the least-squares correlation scaled to the quoted marginal uncertainties; the cross denotes the model prediction. **c,** Sinusoidal power of the full record after projection of segment offsets, before and after subtracting the fitted full model. Power is the squared sinusoidal amplitude divided by two, on a linear axis in units of 10^{-36}. The dotted line is the 99% quantile of the maximum power over 5–40 μHz from 200 resampled residual series. **d,** Sixty-second averages around the excursion on 7 July 2026; red points mark averages consisting mainly of samples excluded by the criterion in Methods. **e,** Distribution of deviations of 300-s means from a running median over ±30 min, divided by the robust standard deviation within each segment. The dashed line marks the threshold of six. The count axis is logarithmic. **f,** Observation time excluded in each campaign (summer in red).

Extended Data Fig. 8 | Radial-displacement validation of the loading convolution. a,b, Amplitude ratios (**a**) and phase differences (**b**) between radial displacements calculated with the loading Green's function and the FES2022 load-tide grid in the centre-of-mass frame, at Shanghai and Wuhan, for eight constituents. The S_2 difference is discussed in Supplementary Methods SM1.

Extended Data Table 1 | Analysis definitions and uncertainty checks.

a, Main and filtered validation analyses. Both reductions use the same 54 observing segments, each with its own offset. They differ in selection, filtering, tidal-template realization and weighting, so their estimates are related but are not independent measurements.

b, Injection tests of the primary estimator. Lunar amplitudes fitted with segment offsets and no empirical daily terms to resampled residuals plus either the full tidal model ($A_{\mathrm{true}} = 1$) or the model without the lunar group ($A_{\mathrm{true}} = 0$). The four daily-disturbance scenarios are defined in Methods, with 300 realizations per row. Bias and standard deviation describe the fitted amplitude. The median standard error is that computed within each realization, before the final residual-based calibration. Coverage is the fraction of intervals, formed with these uncalibrated standard errors, containing the injected amplitude; with the calibrated uncertainty of 0.14 the one-standard-error coverage is 0.67–0.68 (Supplementary Methods SM3); the accompanying 95% binomial interval refers to one-standard-error coverage. The last column gives the fraction with $A > 3\sigma$.

c, Noise-model simulations of the alternative processing. Bias and coverage of nominal 95% intervals based on the largest of the least-squares, delete-day and delete-period standard errors. *T* denotes the full template without daily terms, and *L* the lunar group with daily terms, using the ten-day calendar-baseline design in Extended Data Table 2b and the actual observation times. Gaussian scenarios contain 1,024 simulations; Student-*t* and moving-average scenarios contain 4,096. OU denotes Ornstein–Uhlenbeck noise and MA(1) a first-order moving-average process.

Extended Data Table 2 | Response estimates and related analyses.

a, Daily terms and campaign-specific tidal amplitudes. Amplitude and local time of maximum of the 24-h term, and amplitude of the 12-h term, fitted jointly with the full tidal model within each campaign. Days are local calendar days with data. Daily amplitudes are expressed in 10^{-18}. Their uncertainties are the larger of the scaled least-squares and day-by-day jackknife values; uncertainties in the time of maximum are jackknife estimates. Weak sinusoidal amplitudes are biased positive. The full-template amplitude *A*, fitted without daily terms, and its residual-calibrated uncertainty are those in Extended Data Fig. 1a. The last column gives the lunar amplitude fitted independently to each campaign with the primary estimator.

b, Related tidal estimates. These estimates share observations and are not combined. The last column specifies each uncertainty definition. Applying the segment-wise estimator of Hou et al.[23] to the summer record with the tidal model used here gives 0.78 ± 0.19.

c, Free sinusoidal coefficients in the auxiliary reduction. Joint M_2 and S_2 cosine–sine coefficients, after controlling 54 segment constants and freely phased N_2 and K_2 terms. Coefficients and standard errors are in units of 10^{-18}. Phases refer to 6 September 2025, 16:05 UTC. Scaled least-squares errors use the residual variance; UTC-day jackknife errors delete each of the 43 observed dates and refit all coefficients.

d, Template-response signal-to-noise ratios. The first four rows use the non-overlapping full-template or three-group fits; their empirical errors are the larger of scaled-least-squares and UTC-day-jackknife errors. The final two rows use the equal-weight complete solid-Earth/loading fit with UTC-day-jackknife errors. The common-white-noise calculation retains each estimator and uses $\alpha = 5.4 \times 10^{-16}\sqrt{\text{s}}$. The dimensionless factor η_i is defined in Methods.